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EVALUATION OF WINTER CLIMATE FOR BREAKING BUD REST USING THE DYNAMIC MODEL

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Abstract

A model aimed at explaining the complex effects of temperatures on completion of rest in peach buds was developed. It was termed "the dynamic model" and is unique as it analyzes the direct temperature effects but also considers the modifying effects of temperatures before and after the specific exposure. The model was developed on the basis of former published experimental data and is based on a two step scheme. The first step produces a reversible intermediate and the second one fixes the effect by an irreversible transition. The rate constant of the first step are assumed to depend on temperature according to the Arrhenius law. Mathematical analysis of the dependence of the intermediate level with time reveals a curve that follows an asymptotic function. The higher the temperature the lower the intermediate maximal level. The irreversible transition is assumed to occur only when a critical mass of the intermediate is reached. This occurs only after exposure to low temperatures. Once this transition takes place, an accumulation of one portion occurs and a renewed accumulation of the intermediate may follow. A computer program was developed on the basis of the dynamic model with suggested parameters from a simulated work.

The model agrees well with the following effects: The bell shape dependence on temperature for rest completion; the negation of the chilling effect by high temperatures; the dependence of the negation effect on the cycle length; the dependence of the negation effect on the level and duration of exposure to high temperature; and the enhancement effect of short exposure to moderate temperatures in daily cycles. Predictions of the dynamic model that long diurnal

exposures to moderate temperatures with short chilling intervals will have a promotive effect was verified experimentally.

1.Introduction

Since the early 1970s a few climatic models have been developed in order to quantify the winter chilling hours and relate the meteorological observations to tree phenology. These models were meant to forecast the expected time of bloom and vegetative sprouting for different varieties of trees. A chill unit model counting the effective chilling hours in the winter season and relating the accumulated weighted chilling units to observed rest completion was developed by Richardson et al. [1974] for two peach cultivars and was further expanded to other cultivars and species and is being adapted to lower latitudes [Del Real-Laborde,1986]. A similar rest completion prediction model for a low-chilling cultivar of nectarine was proposed by Gilreath and Buchanan [1981] and for apples by Shaltout and Unrath [1983]. These models are based on the following experimental observations: The temperature response for rest completion follows a bell - shaped function; the contribution to rest completion from each chilling hour is independent of temperatures during the preceding or subsequent periods; and high temperatures have a negative effect which depends on their level. A similar treatment is suggested by Seeley and Damavandy for the response of seeds of deciduous fruits to stratification temperatures [1985].

Experiments under controlled temperature conditions [Couvillon and Erez,1985; Erez and Couvillon, 1987; Erez et al, 1979a, 1979b; Erez and Lavee, 1971], revealed that the effects of temperature on the process of rest breaking in peach buds can not be described by the chill - unit model. In experiments with periodically changed temperatures [Erez et al, 1979b], cycles of 1, 3, 6 and 9 days were examined, with two thirds of the cycle length at 4-6°C and the remaining one third at 24°C. Each cycle was repeated until the plants were exposed to a fixed number of chilling hours. The results show strong dependence of rest breaking on the cycle length: from 100% chilling negation during the warm part of 1-day cycles to no negation for 9-day cycles. According to the chill - unit model, the results had to be the same for all cycles. Furthermore, diurnal cycles with a different ratio of chill (4 - 6°C) to moderate (15 ^{o}C) temperatures [Couvillon and Erez,1985] revealed an effect of chilling enhancement by brief exposures to moderate temperatures. It was also shown that the negating effect of chilling by high temperature depends not only on the temperature's level but also on its duration [Couvillon and Erez,1985]. These

observations were not incorporated in the chill - unit model and therefore can not be predicted by it.

Erez and Couvillon [1987] assumed that the multiple effects of temperature may be explained by means of a two-stage scheme where the first step is a reversible process which has to explain the action of chilling and high temperatures, and the second step is the moderate temperature reaction that fixes the chilling effect. Earlier, Purvis and Gregory used a similar scheme in their study of vernalization of rye plants [1952]. The mathematical analysis of these schems was not performed. Recently Fishman *et al.* [1987a,1987b] suggested and analyzed mathematically a new model termed "the dynamic model". In the present paper experimental observations and the dynamic model predictions are compared and an additional experiment under controlled conditions, designed to test the model predictions, is reported. Details of the model are described and its use is demonstrated.

1.1 The dynamic model - theory

The rest completion is assumed to be dependent on an accumulated amount of some chemical (enzyme) or physical (structure) changes in plants. We have termed this product the rest completion factor (RCF). On the basis of experimental evidence this factor is assumed to have a thermally labile intermediate (IRCF). Following Erez and Couvillon's suggestions [3], a two-step process is thus considered to be responsible for the changes in the buds leading to rest breaking:

$$\text{REST} \underset{k_1}{\overset{k_0}{\rightleftharpoons}} \text{IRCF} \dashrightarrow \text{RCF} \qquad (1)$$

The first step is a reversible process of formation and destruction of IRCF. The second step is suggested to act as a cooperative transition: When a critical level of the intermediate is amassed, a quantum of IRCF is transferred irreversibly to a quantum of stable RCF. The rate constants for formation (k_0) and destruction (k_1) of the IRCF are assumed to obey the Arrhenius law:

$$k_{0,1} = A_{0,1}\exp(-E_{0,1}/T) \qquad (2)$$

where T is temperature in absolute degrees (^{o}K); E_0 and E_1 are energies of activation for formation and destruction processes, respectively, in absolute

temperature units (^{o}K); and A_0 and A_1 are coefficients independent of the temperature. I.e., all temperature effects may be described by using only the parameters E_0, E_1, A_0, and A_1, as was shown before [8]. The simplest dynamic equation adequate to the first step of the scheme (1) is the linear one:

$$db/dt = k_o - k_1 b \qquad (3)$$

where b is the level of the IRCF. The level of RCF is designated as c. If the temperature is constant the solution for b is a smooth curve:

$$b = b_s - (b_s - b_o)\exp(-k_1 t) \qquad (4)$$

where b_s is the steady state level ($b_s = k_o / k_1$) and b_o is the initial value of b. If the steady state is greater than the critical level b_c (Fig.1),when b reaches the level b_c, one portion of b is passed to c, while b drops to zero and the dynamic curve will exhibit a periodic form. If the steady state is below the critical level b_c, the transition from b to c will never occur and rest will not be completed.

Analysis of the solution of eq. (4) shows that, at the early time course, the initial angle of the b curve lies higher, the higher the temperature independent of the parameter values. On the other hand, the level of the asymptotic lines for different temperatures depends on the parameters E_0 and E_1. We assume that $E_1 > E_0$, then the higher the temperature, the lower the steady state will lie (Fig. 2), in agreement with the pattern of behavior of the dormant bud. The dependence of accumulation of portions of 'b' on temperature is not monotonous, but has a maximum. It can be seen from Fig. 3 that the dynamic curve for a certain temperature T_2 crosses the critical level earlier than the dynamic curves for higher (T_3) as well as for lower (T_1) temperatures. This means that more portions of RCF will be accumulated in the same period of time under T_2 than under the higher (T_3) or the lower (T_1) temperatures. It is important to note that according to the dynamic model, any change of temperature will cause a shift from one steady state to another. This explains the negation of the chilling effect by high temperatures. Comparing cycles of low with high temperatures shows (Fig. 4) that in short cycles when the critical level has not been reached by the cold portion of the cycle, the accumulated b level drops when temperature is elevated, to follow the dynamic steady state of its temperature. Analysis of the resulting curve shows that as this combined asymptote is lower than the critical level, no RCF is

accumulated and hence under this special temperarture cycle, no rest breaking is obtained, in spite of a long exposure to chilling temperatures. When a moderate temperature, T_2, is applied during time t_1 (Fig. 5) and then the temperature changes to the chill one, T_1, the chilling regime starts now from the point $b = b_1$ which may be higher than the respective point in the case of continuous chilling (b_2). The dynamic curve for T_2 does not intersect with the critical line b_c , which means that continuous application of this temperature results in zero accumulation of RCF and hence in no rest breaking. The curve for T_1 intersects with the critical level at time t_3, which means that the time t_3 is needed for production of one portion of RCF. The combined dynamic curve T_3 crosses the critical line at t_2 earlier than the curve of the continuous chilling, which means that more portions of RCF are produced during the same time in the combined regime than with the continuous chilling. The enhancing effect is amplified when the length of the cycle is close to the time needed for achieving the critical level and when the exposure to moderate temperature is short enough. During the cycle, longer exposure period to moderate temperature leads to a reduced chilling effect, as the respective dynamic curve crosses the critical line later (Fig. 5 at t_4) than the curve for the continuous chilling (Fig. 5 at t_3). However, per unit of chilling, this is the most efficient combination.

1.2. The dynamic model - parameters.

The mathematical analysis done on the dynamic model [8] related the theoretical parameters with the following experimentally measurable values: (i) The temperature leading to a maximal chilling effect was taken as 7°C; (ii) the highest temperature which still gives a positive response was taken as 13°C (T_c); and (iii) the time needed to accumulate one critical portion of RCF under optimal temperature conditions was assumed to be 28 hrs. The relationships were reported previously [8]. The equations are nonlinear but can be solved numerically. The numerical computations led to the following set of parameters: $E_0 = 0.4153 \times 10^4$, $E_1 = 0.1289 \times 10^5$, $A_0 = 0.1395 \times 10^6$, $A_1 = 0.2567 \times 10^{19}$.

2. Materials and methods

Cv. "Desertgold" peach hardwood cuttings were collected from The Volcani Center farm at Bet Dagan, Israel, in January 1985 and rooted as described previously [7]. The plants were grown in 5-l containers filled with vulcanic tuff and peat moss and fertigated regularly. The plants were taken for the experiment on

November 15 before any chilling was recorded outdoors. Two Conviron E7H temperature-controlled chambers were set for 6/14°C daily cycles with ratios of 8/16 and 12/12 hrs. A continuous 6°C condition was used for comparison. Light (50 μmol s^{-1} m^{-2}) was supplied during the warmer period. The plants were exposed to these regimes for 40 days. Five replicates of single plants per treatment were used.

Following chilling exposure the plants were placed in the laboratory at 22°C for forcing in light (50 μmol s^{-1} m^{-2}). Bud break (green tip stage for vegetative buds and petal appearance in floral buds) counts were recorde weekly for 8 weeks.

3. Results

On the basis of the theoretical model and the parameters estimated as stated above, a computer simulation model was developed (see details in Appendix 1). The dynamic model predicted that extreme promotion of rest completion per unit of chilling will occur if long periods of moderate (14°C) temperatures are cycled with short periods of chilling (6°C) in a daily cycle (Fig. 5). This was tested with cv. "Desertgold". Similar levels of vegetative as well as floral bud break were obtained with 320, 480 and 960 chilling hrs (Fig. 9). From 65.2% to 86.3% and from 91.6% to 97.3% with vegetative and floral buds respectively. The two earlier treatments had 16 and 12 hrs daily at 14°C. This effect was not due to saturation of the chilling requirement by the smaller amount of chilling, since in a former experiment 850 hrs at continuous 6oC resulted in only 21% vegetative bud break. The prediction of the dynamic model follows a similar trend as against the Utah model.

Climatic data from Caçador SC Brazil in winters 1984 and 1985 were run with the computer programs of the dynamic and the Utah models and compared to phenological observation. A comparison between the dynamic and the Utah models is seen in Fig. 7. 1984 was a cooler year and bud break especially of vegetative buds was better than in 1985 in peaches and apples. This was not the case with the lower chilling requiring peach flower buds and apple spur mixed buds. Both models show the advantage of 1984 over 1985 but with relative quantitative difference. A more detailed- monthly analysis reveal stronger variations (fig. 8). Further field experiments are needed.

4. Discussion

The dynamic model was developed using experimental results and is based on a coherent theory trying to cope with the hitherto unexplained mode of effects of

temperature on rest development in buds. It shows a close fit with former experimental data obtained under controlled temperature conditions and predicted correctly the response of peach buds to specific regimes of temperature combinations. The bell-shaped curve is accepted by the Utah chill-unit model as a hypothesis, whereas in the dynamic model this curve is calculated using basic assumptions. The changes in the effect of high temperature as a function of temperature level, of the cycle length, and of the duration in the daily cycle, are explained by the dynamic characteristic of the model. By this we mean dependence of the response to temperature on former and subsequent conditions. In this respect our model is the only one that considers this critical point.

Another interesting development is the reappearance of a threshold or "critical temperature." This is the highest temperature that will allow the accumulation of a critical mass of IRCF. It seems to lie between 14 and 13°C This temperature has to be distinguished from the 7.2°C threshold that was suggested by Weinberger for calculating accumulation of chilling hrs and is commonly used for assessing chilling requirements of deciduous fruit tree cultivars [Weinberger 1956].

The promotive effect of moderate temperatures cycling with chilling is explained by the dynamic model by accelerating the accumulation of RCF. This will happen only at certain temperatures which are above but close to the critical temperature.
The dynamic model also stresses the promotive effect of cycling low/moderate temperatures. The data presented in Figure 9 point to a similar effect on breaking rest per hour of chilling, and of moderate temperature when cycled in a daily cycle in the proper ratio. This finding has enormous weight for evaluating the effect of temperature regime on rest completion, especially under mild winters. All this point to the over-simplification of a method counting just the accumulation of positive chilling hours and negative high temperature hours, ignoring the dynamic mutual effects between temperatures and the change in response due to them.

The computer program detailed in Appendix 1 may be used for analyzing correctly the effects of the climatic conditions on dormant buds by the dynamic model. For cv. "Redhaven" vegetative buds we consider good bud break (>50%) when 75 portions have accumulated; for cv. "Desertgold" the value is 50 portions. Although our work was confined to peach buds only, we believe the framework of the model applies also to the apple and to other Rosaceae fruit tree species. This assumption is based on the similarity of the response between the apple [Shaltout

and Unrath, 1983] and the peach [Erez and Couvillon, 1987; Gilreath and Buchanan, 1981].

The model presented here seems to follow closely the actual response of the dormant bud to temperature. This means that the factors controlling rest respond to temperature in a manner described by the two-step model, with the first step being reversible. It is hoped that this may lead to a new approach to the investigation of the mechanism of rest in buds.

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Appendix 1 - computer program

```
program cycles
dimension tet(250)
 open(unit=1,file='enhanjnd.dat',status='old')
 open(unit=2,file='cycljnd.lis',status='new')
n = 0
e0= 0.4153e+04
e1= 0.1289e+05
a0= 0.1395e+06
a1= 0.2567e+19
e  = e1 - e0
tetmlt = 277.
```

```
      slp = 1.6
      do while (n .eq. 0)
      read(1,*,end=50) np,teta,teta1,nwarm,nhr
      x0 = 0.
      y  = 0.
     do 5 l = 1,nhr
 5   tet(l) = teta
     do 7 l = 1,nwarm
 7   tet(l) = teta1
     do 20 j = 1,np
     do 10 i = 1,nhr
     ftmprt = slp*tetmlt*(tet(i) - tetmlt) / tet(i)
     sr = exp(ftmprt)
     xi = sr / (1. + sr)
     xs = (a0/a1)*exp(e/tet(i))
     ak1= a1*exp(-e1/tet(i))
     x  = xs - (xs-x0)*exp(-ak1)
     x0 = x
     if (x.ge.1.) then
     delt = x*xi
     x0 = x - delt
     y  = y + delt
     end if
 10  continue
 20  continue
     write(2,40) y,teta1, nwarm
 40   format (1x,f5.1,3x,f5.1,3x,i5)
    end do
 50  stop
    end
```

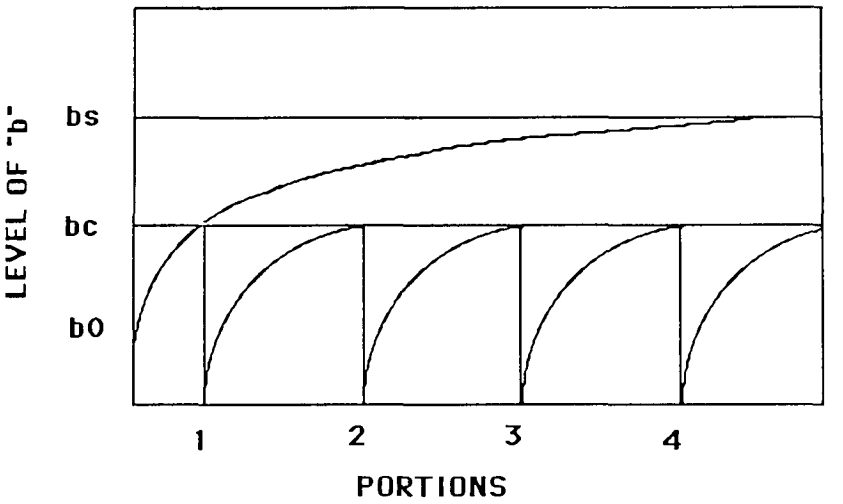


Fig.1: The effect of continuous chilling at a constant low temperature on portion accumulation of the rest breaking factor 'b'.

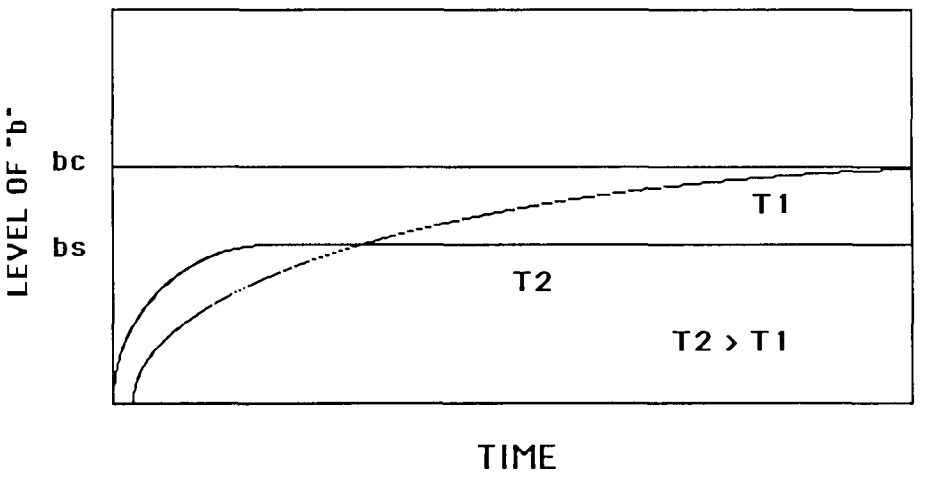


Fig. 2: The effect of temperature on the dynamics of the accumulation of the intermediate rest breaking factor (T1- chilling temperature; T2- nonchilling temperature).

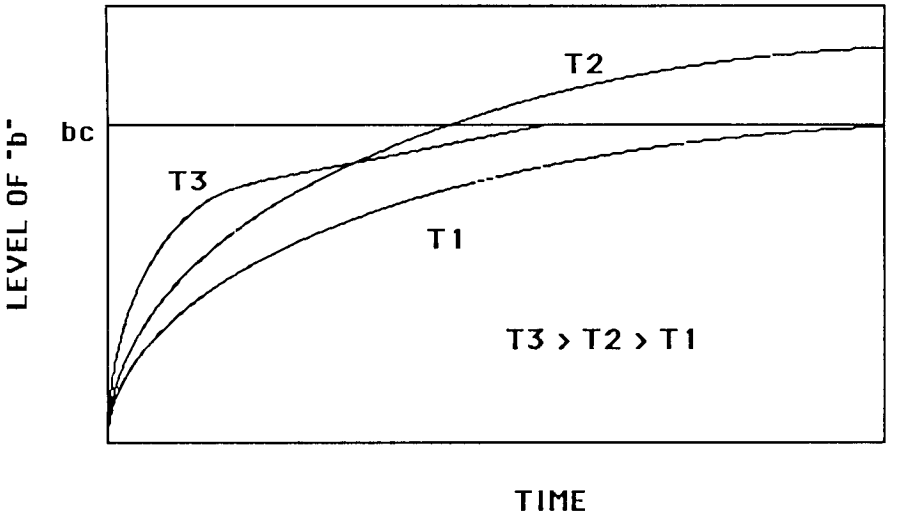


Fig.3: The relative efficiency of various temperatures on the dynamics of the accumulation of the intermediate rest breaking factor 'b'. (T2- optimum chilling temperature).

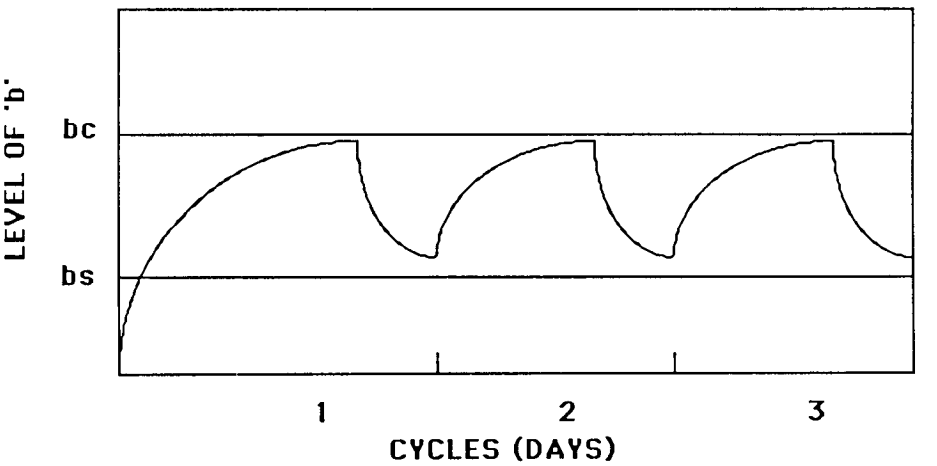


Fig. 4: The effect of alternating low (chilling) and high (negating) temperatures in a diurnal cycle on the accumulation of the intermediate rest breaking factor-'b'. (bc- critical level of 'b'; bs- steady dtate for the high temperature).

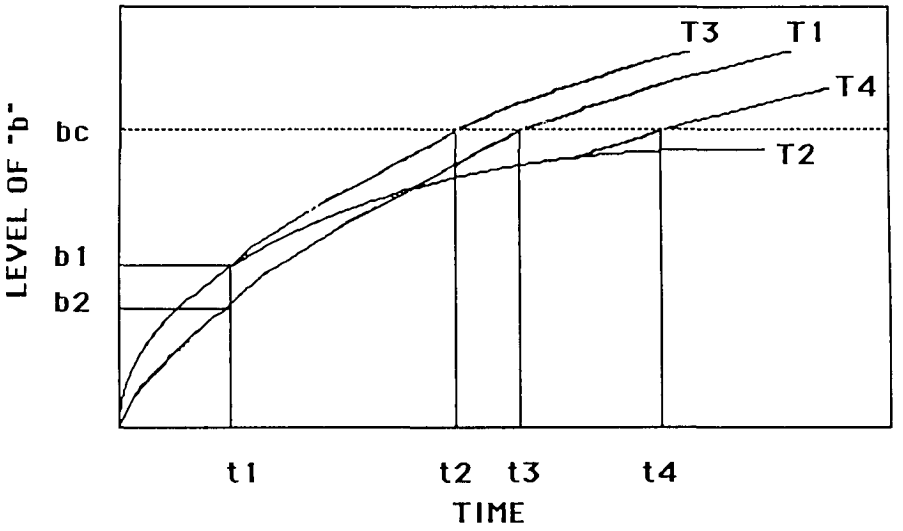


Fig.5: The effect of moderate temperature alternating with chilling on the dynamics of the accumulation of the intermediate rest breaking factor 'b'.

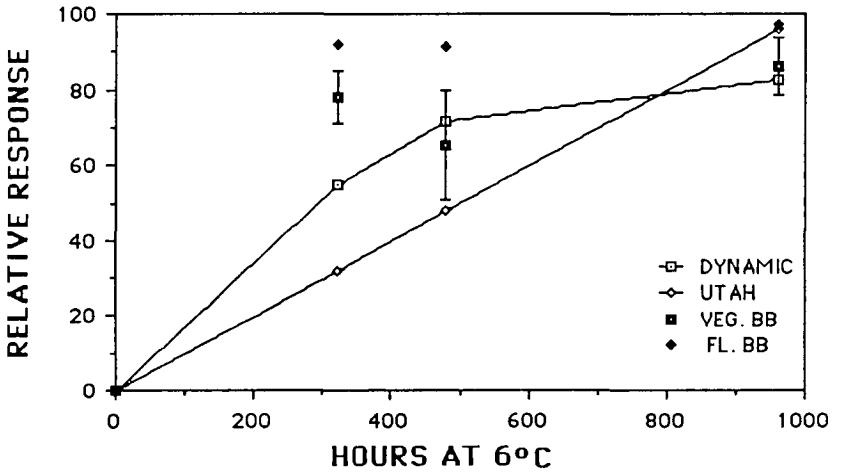


Fig. 6: The effect of different periods of exposure to chilling on vegetative(VEG BB) and floral(FL BB) bud break in 'Desertgold' peach with prediction of the dynamic and Utah models.(960 hrs at daily cycles of 6/14°C - 8:16 hrs; 12:12 hrs and 24:0 hrs).

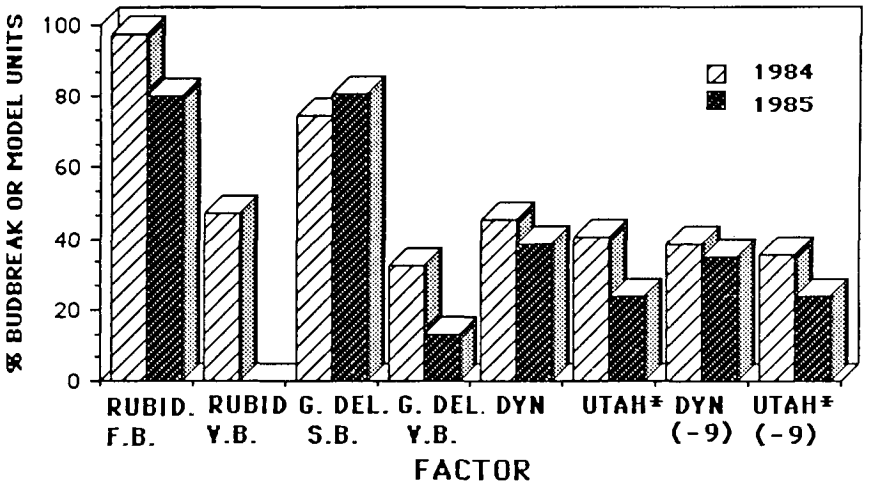


Fig. 7: Budbreak of Rubidoux peach and Golden Delicious apple in Caçador SC Brazil in 1984 and 1985 in comparison to chilling accumulation according to the Utah or the dynamic models. (* Chill units divided by 10; -9 - less values of September)

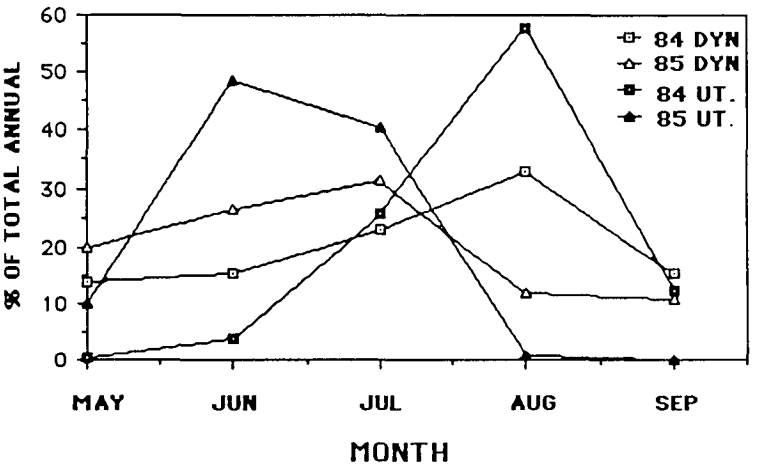


Fig. 8: Monthly accumulation of chill units according to the Utah model or portions according to the dynamic model during 1984 and 1985 in Caçador Brazil.